

MTH 131 Practice Final Exam
Fall 2012

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Solve and check the linear equation.

1) $7x - 9 = -10 + 4x$

1) _____

A) $\left\{-\frac{1}{3}\right\}$

B) $\{-3\}$

C) $\left\{-\frac{11}{19}\right\}$

D) $\{3\}$

Solve the equation.

2) $\frac{x}{6} = \frac{x}{8} + 5$

2) _____

A) $\{30\}$

B) $\{40\}$

C) $\{120\}$

D) $\{48\}$

First, write the value(s) that make the denominator(s) zero. Then solve the equation.

3) $\frac{x-8}{2x} + 4 = \frac{x+3}{x}$

3) _____

A) No restrictions; $\left\{\frac{11}{8}\right\}$

B) $x \neq 0, 2; \{2\}$

C) $x \neq 0; \{-10\}$

D) $x \neq 0; \{2\}$

Solve the equation.

4) $\frac{9}{y+2} - \frac{3}{y-2} = \frac{12}{y^2-4}$

4) _____

A) $\{\sqrt{19}\}$

B) $\{6\}$

C) $\{-6\}$

D) $\{36\}$

5) $\frac{m+4}{m^2+9m+20} - \frac{4}{m^2+10m+25} = \frac{m-4}{m^2+9m+20}$

5) _____

A) $\{6\}$

B) $\{12\}$

C) $\{-6\}$

D) $\{-24\}$

Solve the problem.

- 6) A local race for charity has taken place since 1993. Using the actual speeds of the winners from 1993 through 1998, mathematicians obtained the formula $y = 0.18x + 5.5$, in which x represents the number of years after 1993 and y represents the winning speed in miles per hour. In what year is the winning speed predicted to be 7.48 mph?

6) _____

A) 2006

B) 2003

C) 2005

D) 2004

- 7) You inherit \$70,000 from a very wealthy grandparent, with the stipulation that for the first year, the money must be invested in two stocks paying 4% and 10% annual interest, respectively. How much should be invested at each rate if the total interest earned for the year is to be \$4000?

7) _____

A) \$20,000 invested at 4%; \$50,000 invested at 10%

B) \$40,000 invested at 4%; \$30,000 invested at 10%

C) \$50,000 invested at 4%; \$20,000 invested at 10%

D) \$30,000 invested at 4%; \$40,000 invested at 10%

- 8) A car rental agency charges \$150 per week plus \$0.20 per mile to rent a car. How many miles can you travel in one week for \$250? 8) _____
- A) 200 miles B) 500 miles C) 1250 miles D) 475 miles
- 9) After a 14% price reduction, a boat sold for \$28,380. What was the boat's price before the reduction? (Round to the nearest cent, if necessary.) 9) _____
- A) \$3973.20 B) \$32,353.20 C) \$202,714.29 D) \$33,000

Solve the formula for the specified variable.

- 10) $F = \frac{9}{5}C + 32$ for C 10) _____
- A) $C = \frac{9}{5}(F - 32)$ B) $C = \frac{F - 32}{9}$ C) $C = \frac{5}{9}(F - 32)$ D) $C = \frac{5}{F - 32}$
- 11) $A = \frac{1}{2}h(b_1 + b_2)$ for b_1 11) _____
- A) $b_1 = \frac{A - h(b_2)}{2h}$ B) $b_1 = \frac{2A - (h)(b_2)}{h}$
- C) $b_1 = \frac{h(b_2) - 2A}{h}$ D) $b_1 = \frac{(b_2)^2 A - h}{h}$
- 12) $R = nE - nr$, for n 12) _____
- A) $n = R - E + r$ B) $n = R + nr - E$ C) $n = \frac{R + nr}{E}$ D) $n = \frac{R}{E - r}$

Solve the equation by factoring.

- 13) $x^2 = x + 6$ 13) _____
- A) $\{-2, 3\}$ B) $\{2, 3\}$ C) $\{-2, -3\}$ D) $\{1, 6\}$
- 14) $15x^2 + 26x + 8 = 0$ 14) _____
- A) $\left\{\frac{4}{3}, -\frac{2}{5}\right\}$ B) $\left\{\frac{4}{3}, \frac{2}{5}\right\}$ C) $\left\{-\frac{4}{3}, -\frac{2}{5}\right\}$ D) $\left\{-\frac{4}{15}, -\frac{1}{4}\right\}$

Solve the equation by the square root property.

- 15) $2(x - 6)^2 = 10$ 15) _____
- A) $\{-11, -1\}$ B) $\{6 \pm \sqrt{5}\}$ C) $\{-6 \pm \sqrt{5}\}$ D) $\{1, 11\}$

Solve the equation using the quadratic formula.

- 16) $3x^2 + 10x + 4 = 0$ 16) _____
- A) $\left\{\frac{-5 - \sqrt{13}}{6}, \frac{-5 + \sqrt{13}}{6}\right\}$ B) $\left\{\frac{-5 - \sqrt{13}}{3}, \frac{-5 + \sqrt{13}}{3}\right\}$
- C) $\left\{\frac{-10 - \sqrt{13}}{3}, \frac{-10 + \sqrt{13}}{3}\right\}$ D) $\left\{\frac{-5 - \sqrt{37}}{3}, \frac{-5 + \sqrt{37}}{3}\right\}$

Solve the polynomial equation by factoring and then using the zero product principle.

17) $3x^4 - 75x^2 = 0$

A) $\{-5, 0, 5\}$

B) $\{-5, 5\}$

C) $\{0\}$

D) $\{-5\sqrt{3}, 0, 5\sqrt{3}\}$

17) _____

18) $4x - 5 = 16x^3 - 20x^2$

A) $\left\{0, \frac{5}{4}\right\}$

B) $\left\{-\frac{1}{4}, \frac{1}{4}, \frac{5}{4}\right\}$

C) $\left\{-\frac{1}{2}, \frac{1}{2}, \frac{4}{5}\right\}$

D) $\left\{-\frac{1}{2}, \frac{1}{2}, \frac{5}{4}\right\}$

18) _____

Solve the radical equation, and check all proposed solutions.

19) $\sqrt{x+1} = 3$

A) $\{10\}$

B) $\{16\}$

C) $\{9\}$

D) $\{8\}$

19) _____

20) $x - \sqrt{3x-2} = 4$

A) $\{9\}$

B) $\{1, 2\}$

C) $\{2, 9\}$

D) $\{-1\}$

20) _____

Solve the absolute value equation or indicate that the equation has no solution.

21) $|x - 1| = 3$

A) $\{2, 4\}$

B) $\{-2, 4\}$

C) $\{-4\}$

D) $\emptyset$

21) _____

22) $3|x - 3| = 18$

A) $\{3, -9\}$

B) $\{9, -3\}$

C) $\{3\}$

D) $\emptyset$

22) _____

Determine whether the relation is a function.

23) $\{(-3, -3), (3, -4), (6, 2), (9, -4), (10, -4)\}$

A) Not a function

B) Function

23) _____

24) $\{(-7, -1), (-7, 2), (-1, 8), (3, 3), (10, -7)\}$

A) Function

B) Not a function

24) _____

Evaluate the function at the given value of the independent variable and simplify.

25) $f(x) = x^2 - 1$; $f(x - 2)$

A) $x^2 - 4x + 4$

B) $x^2 - 4x + 3$

C) $x^2 + 4$

D) $x^2 - 3$

25) _____

Evaluate the piecewise function at the given value of the independent variable.

26) $f(x) = \begin{cases} x - 3 & \text{if } x > -5 \\ -(x - 3) & \text{if } x \leq -5 \end{cases}$; $f(-8)$

A) 11

B) -11

C) 20

D) -8

26) _____

27) $h(x) = \begin{cases} \frac{x^2 - 3}{x + 5} & \text{if } x \neq -5 \\ x - 1 & \text{if } x = -5 \end{cases}$; $h(-5)$

A) -6

B) 6

C) undefined

D) -4

27) _____

Find the slope of the line that goes through the given points.

28) $(3, -5), (-9, -8)$

A) $\frac{1}{4}$

B) $-\frac{1}{4}$

C) 4

D) $\frac{13}{6}$

28) _____

29) $(-1, 4), (5, 4)$

A) 0

B) $\frac{4}{3}$

C) 2

D) Undefined

29) _____

Use the given conditions to write an equation for the line in point-slope form.

30) Slope = 4, passing through $(-3, 7)$

A) $x - 7 = 4(y + 3)$

B) $y - 7 = 4(x + 3)$

C) $y + 7 = 4(x - 3)$

D) $y = 4x + 19$

30) _____

Use the given conditions to write an equation for the line in slope-intercept form.

31) Slope = -4, passing through $(8, 4)$

A) $y - 4 = -4x - 8$

B) $y - 4 = x - 8$

C) $y = -4x + 36$

D) $y = -4x - 36$

31) _____

32) Passing through $(4, 8)$ and $(7, 3)$

A) $y = mx + \frac{44}{3}$

B) $y - 8 = -\frac{5}{3}(x - 4)$

C) $y = -\frac{5}{3}x + \frac{44}{3}$

D) $y = \frac{5}{3}x + \frac{44}{3}$

32) _____

Determine the slope and the y-intercept of the graph of the equation.

33) $y - 8 = 0$

A) $m = 0$; no y-intercept

B) $m = 1$; $(0, 8)$

C) $m = 0$; $(0, 8)$

D) $m = 8$; $(0, 0)$

33) _____

34) $7x - 10y - 70 = 0$

A) $m = 7$; $(0, 70)$

B) $m = \frac{10}{7}$; $(0, 10)$

C) $m = \frac{7}{10}$; $(0, -7)$

D) $m = -\frac{7}{10}$; $(0, 7)$

34) _____

Solve.

35) The average value of a certain type of automobile was \$14,280 in 1992 and depreciated to \$5460 in 1997. Let y be the average value of the automobile in the year x , where $x = 0$ represents 1992. Write a linear equation that models the value of the automobile in terms of the year x .

35) _____

A) $y = -1764x + 5460$

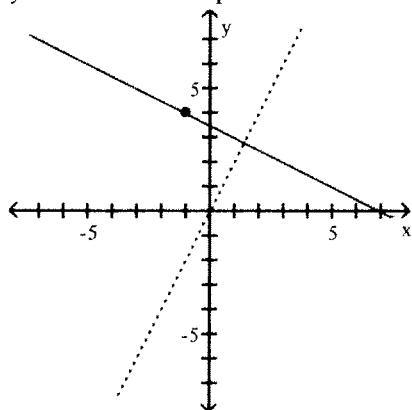
B) $y = -1764x - 3360$

C) $y = -1764x + 14,280$

D) $y = -\frac{1}{1764}x - 5460$

Find an equation for the line with the given properties.

- 36) The solid line L contains the point $(-1, 4)$ and is perpendicular to the dotted line whose equation is $y = 2x$. Give the equation of line L in slope-intercept form. 36) _____



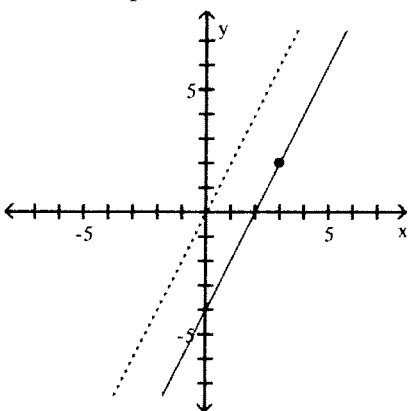
A) $y = \frac{1}{2}x + \frac{7}{2}$

B) $y = -\frac{1}{2}x + \frac{7}{2}$

C) $y - 4 = 2(x + 1)$

D) $y - 4 = -\frac{1}{2}(x + 1)$

- 37) The solid line L contains the point $(3, 2)$ and is parallel to the dotted line whose equation is $y = 2x$. Give the equation for the line L in slope-intercept form. 37) _____



A) $y = 2x - 1$

B) $y - 2 = 2(x - 3)$

C) $y = 2x - 4$

D) $y = 2x + b$

Use the given conditions to write an equation for the line in the indicated form.

- 38) Passing through $(2, 5)$ and perpendicular to the line whose equation is $y = \frac{1}{6}x + 8$; 38) _____

slope-intercept form

A) $y = -6x - 17$

B) $y = -\frac{1}{6}x - \frac{17}{6}$

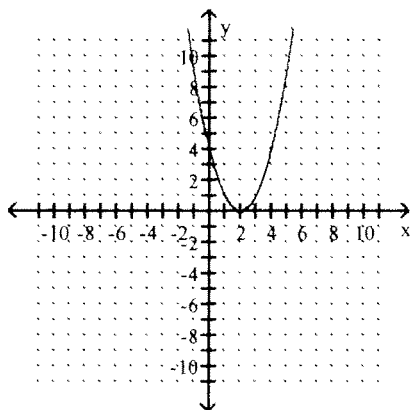
C) $y = 6x - 17$

D) $y = -6x + 17$

The graph of a quadratic function is given. Determine the function's equation.

39)

39) _____



A) $h(x) = x^2 - 2$

B) $j(x) = x^2 + 2$

C) $f(x) = x^2 - 4x + 4$

D) $g(x) = x^2 + 4x + 4$

Determine whether the given quadratic function has a minimum value or maximum value. Then find the coordinates of the minimum or maximum point.

40) $f(x) = x^2 + 2x - 9$

40) _____

A) maximum; $\{-1, -10\}$

B) minimum; $\{-10, -1\}$

C) maximum; $\{-10, -1\}$

D) minimum; $\{-1, -10\}$

Solve the problem.

41) The cost in millions of dollars for a company to manufacture x thousand automobiles is given by the function $C(x) = 3x^2 - 18x + 63$. Find the number of automobiles that must be produced to minimize the cost.

41) _____

A) 3 thousand automobiles

B) 9 thousand automobiles

C) 36 thousand automobiles

D) 6 thousand automobiles

Determine whether the function is a polynomial function.

42) $f(x) = \pi x^3 - 4x^2 - 5$

42) _____

A) Yes

B) No

43) $f(x) = x^{3/2} - x^6 + 6$

43) _____

A) Yes

B) No

44) $f(x) = \frac{7 - x^2}{2}$

44) _____

A) Yes

B) No

Find the x-intercepts of the polynomial function. State whether the graph crosses the x-axis, or touches the x-axis and turns around, at each intercept.

45) $f(x) = 8x^2 - x^3$

- | | |
|--|--|
| A) 0, crosses the x-axis; | B) 0, touches the x-axis and turns around; |
| $2\sqrt{2}$, crosses the x-axis; | $2\sqrt{2}$, crosses the x-axis; |
| $-2\sqrt{2}$, crosses the x-axis | $-2\sqrt{2}$, crosses the x-axis |
| C) 0, touches the x-axis and turns around; | D) 0, touches the x-axis and turns around; |
| 8, touches the x-axis and turns around | 8, crosses the x-axis |

45) _____

Use the Leading Coefficient Test to determine the end behavior of the polynomial function.

46) $f(x) = -3x^4 + 4x^3 - 5x^2 - 4x - 1$

- | | |
|---|---|
| A) falls to the left and falls to the right | B) rises to the left and rises to the right |
| C) falls to the left and rises to the right | D) rises to the left and falls to the right |

46) _____

47) $f(x) = 5x^3 + 4x^2 - 5x + 5$

- | | |
|---|---|
| A) falls to the left and rises to the right | B) rises to the left and falls to the right |
| C) rises to the left and rises to the right | D) falls to the left and falls to the right |

47) _____

Use the compound interest formulas $A = P\left(1 + \frac{r}{n}\right)^{nt}$ and $A = Pe^{rt}$ to solve.

48) Find the accumulated value of an investment of \$800 at 12% compounded quarterly for 2 years.

- | | | | |
|-------------|--------------|-------------|--------------|
| A) \$992.00 | B) \$1003.52 | C) \$848.72 | D) \$1013.42 |
|-------------|--------------|-------------|--------------|

48) _____

49) Find the accumulated value of an investment of \$7000 at 7% compounded continuously for 6 years.

- | | | | |
|----------------|----------------|----------------|--------------|
| A) \$10,505.11 | B) \$10,753.73 | C) \$10,653.73 | D) \$9940.00 |
|----------------|----------------|----------------|--------------|

49) _____

50) Suppose that you have \$11,000 to invest. Which investment yields the greater return over 5 years: 5.4% compounded monthly or 5.5% compounded quarterly?

- | |
|---|
| A) Both investment plans yield the same return. |
| B) \$11,000 invested at 5.5% compounded quarterly over 5 years yields the greater return. |
| C) \$11,000 invested at 5.4% compounded monthly over 5 years yields the greater return. |

50) _____

Write the equation in its equivalent exponential form.

51) $\log_4 x = 3$

- | | | | |
|--------------|--------------|--------------|--------------|
| A) $3^4 = x$ | B) $4^x = 3$ | C) $4^3 = x$ | D) $x^3 = 4$ |
|--------------|--------------|--------------|--------------|

51) _____

Write the equation in its equivalent logarithmic form.

52) $6^2 = x$

- | | | | |
|-------------------|-------------------|-------------------|-------------------|
| A) $\log_6 x = 2$ | B) $\log_6 2 = x$ | C) $\log_2 x = 6$ | D) $\log_x 6 = 2$ |
|-------------------|-------------------|-------------------|-------------------|

52) _____

53) $\sqrt[3]{64} = 4$

- | | | | |
|--------------------------------|--------------------|--------------------------------|------------------------------|
| A) $\log_{64} 4 = \frac{1}{3}$ | B) $\log_4 64 = 3$ | C) $\log_{64} 3 = \frac{1}{4}$ | D) $\log_4 64 = \frac{1}{3}$ |
|--------------------------------|--------------------|--------------------------------|------------------------------|

53) _____

Evaluate the expression without using a calculator.

54) $2^{\log_2 18}$ 54) _____
 A) 2 B) $\log_2 18$ C) 18 D) 20

55) $\log_3 3^{14}$ 55) _____
 A) 17 B) 3 C) $\log_3 14$ D) 14

Find the domain of the logarithmic function.

56) $f(x) = \log_7 (x - 6)$ 56) _____
 A) $(-6, \infty)$ B) $(-\infty, 6)$ or $(6, \infty)$ C) $(6, \infty)$ D) $(-\infty, 0)$ or $(0, \infty)$

57) $f(x) = \ln (8 - x)$ 57) _____
 A) $(-\infty, 8)$ or $(8, \infty)$ B) $(-\infty, 8)$ C) $(-8, \infty)$ D) $(-\infty, 0)$

Evaluate or simplify the expression without using a calculator.

58) $\log 0.01$ 58) _____
 A) -2 B) 2 C) $\frac{1}{2}$ D) $-\frac{1}{2}$

Use properties of logarithms to expand the logarithmic expression as much as possible. Where possible, evaluate logarithmic expressions without using a calculator.

59) $\log_b (yz^5)$ 59) _____
 A) $5 \log_b yz$ B) $5 \log_b y + 5 \log_b z$
 C) $\log_b y + \log_b 5z$ D) $\log_b y + 5 \log_b z$

60) $\log_6 \left(\frac{7 \cdot 11}{5} \right)$ 60) _____
 A) $\log_6 \left(\frac{77}{5} \right)$ B) $\log_6 7 + \log_6 11 - \log_6 5$
 C) $\log_6 77 - \log_6 5$ D) $\log_6 13$

Use properties of logarithms to condense the logarithmic expression. Write the expression as a single logarithm whose coefficient is 1. Where possible, evaluate logarithmic expressions.

61) $3 \log_y 4 + \log_y 3$ 61) _____
 A) $\log_y 36$ B) $\log_y 192$ C) $\log_y 4$ D) $3 \log_y 12$

62) $6 \ln x - \frac{1}{4} \ln y$ 62) _____
 A) $\ln x^6 y^4$ B) $\ln x^6 \sqrt[4]{y}$ C) $\ln \frac{x^6}{\sqrt[4]{y}}$ D) $\ln \frac{x^6}{y^4}$

Use common logarithms or natural logarithms and a calculator to evaluate to four decimal places

63) $\log_{24} 386$

A) 0.5336

B) 1.2064

C) 1.8741

D) 3.9668

63) _____

64) $\log_{\pi} 16$

A) 2.4220

B) 1.7013

C) 0.4129

D) 0.7070

64) _____

Answer Key

Testname: MTH 131 PRACTICE FINAL EXAM FALL 2011

- 1) A
- 2) C
- 3) D
- 4) B
- 5) C
- 6) D
- 7) C
- 8) B
- 9) D
- 10) C
- 11) B
- 12) D
- 13) A
- 14) C
- 15) B
- 16) B
- 17) A
- 18) D
- 19) D
- 20) A
- 21) B
- 22) B
- 23) B
- 24) B
- 25) B
- 26) A
- 27) A
- 28) A
- 29) A
- 30) B
- 31) C
- 32) C
- 33) C
- 34) C
- 35) C
- 36) B
- 37) C
- 38) D
- 39) C
- 40) D
- 41) A
- 42) A
- 43) B
- 44) A
- 45) D
- 46) A
- 47) A
- 48) D
- 49) C
- 50) B

Answer Key

Testname: MTH 131 PRACTICE FINAL EXAM FALL 2011

- 51) C
- 52) A
- 53) A
- 54) C
- 55) D
- 56) C
- 57) B
- 58) A
- 59) D
- 60) B
- 61) B
- 62) C
- 63) C
- 64) A